

A SSH server for remote administration

Since VirtualBox does not have a dedicated remote administration tool, we will have to SSH into the machine and use *VBoxManage*. So, we need to set up an SSH server in our server.

First install the OpenSSH server:

```
$ sudo apt-get install openssh-server
```

Now open the `/etc/ssh/sshd_config` file and make a few changes: Set the port to something non-standard, such as 99; set `RSAAuthentication` to `Yes`; set `Protocol` to 2; `PasswordAuthentication` to `NO` (we will set up certification-based authentication); and `PermitRootLogin` to `Yes`.

Now we need to set up Public Key authentication. SSH keys are machine-specific, not user specific. So you need to set up one key on each machine that you want to access the server from. From a terminal on a Linux client, type:

```
$ sudo mkdir ~/.ssh && chmod 700 ~/.ssh
$ sudo ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsa
```

Supply a really big passphrase. It can include anything. Just remember this 30 character (optimal) long sentence.

Then secure the files:

```
$ sudo chmod go-w ~/.ssh
$ sudo chmod 700 ~/.ssh
$ sudo chmod go-rwx ~/.ssh/*
```

You subsequently need to change two lines in the client `/etc/ssh/ssh_config` file:

- Set `IdentityFile` to `~/.ssh/id_rsa`
- Set `Protocol` to 2

Repeat this for every Linux client.

In your clients, there will be a `~/.ssh/id_rsa.pub` file. This is the public key file. Copy these files, one by one, to your server and append them to the `~/.ssh/authorized_keys` file (use the `cat` command with the `>>` directive).

Change the server configuration a bit; in the `sshd_config` file set:

- `Protocol` to 2
- `PubKeyAuthentication` to `YES`
- `AuthorizedKeysFile` to `~/.ssh/authorized_keys`
- `PasswordAuthentication` to `no`
- `ChallengeResponseAuthentication` to `no`
- `UsePAM` to `no`

Now do a `/etc/init.d/ssh` restart and you should have a fully working SSH environment. Try to log in, with the `"PreferredAuthentications=publickey"` flag to ssh. You should have it.

For more, consult sial.org/howto/openssh/publickey-auth.

Appendix A: RDP client on Linux

The RDP client on Linux is an X application known as `rdesktop`. To install it, use:

```
$ sudo apt-get install rdesktop
```

...or, on your CentOS VMs:

```
# yum install rdesktop
```

You can connect with one simple command:

```
$ rdesktop server.officenet.local:3390 # adjust the hostname and port.
```

For more information, see the man page.

Appendix B: SSHing from Windows

You use a free program called PuTTY for Telnet and SSH on Windows. PuTTY can be downloaded from tartarus.org/~simon/putty-snapshots/x86/putty-installer.exe.

Install this snapshot. There are a few guides to using PuTTY on the Internet. Use the following two websites to work out how to configure Public Key authentication on PuTTY.

- www.ualberta.ca/CNS/RESEARCH/LinuxClusters/pka-putty.html
 - searchenterprise-linux.techtarget.com/tip/0,289483,sid39_gci1272986,00.html
- Use a 4096 bit SSH2-RSA key pair.

Some notes

1. You must specify a separate VRDP port for every VM. You cannot remote access VMs by name. They must be accessed by port. So maintain a list of names to port number assignments, somewhere.
2. You can use SSH to launch an X application remotely also (you need to use Linux here). Use this link: nixcraft.com/getting-started-tutorials/170-run-remote-x-applications-over-network-using-ssh.html.
3. This one is for creating purely random passphrases and passwords by rolling a dice a number of times: world.std.com/~reinhold/diceware.html. This is highly recommended for security. 

By: Boudhayan Gupta

Boudhayan is a 14-year old student who suffers from an acute psychological disorder called *Distromania*. He owes his life to Larry Page and Sergei Brin. Apart from that, he enjoys both reading and writing, and when he is not playing with his Python :-), during most of his spare time he can be found listening to Fort Minor, or cooking.